

ML Internship Handbook - Part 1: Data Preprocessing & Feature Engineering

EDA (Univariate Analysis)

EDA is the process of understanding a dataset before building models. Univariate analysis focuses on one variable at a time.

Key checks include mean, median, mode, minimum, maximum, variance, standard deviation, skewness, and outliers.

For categorical data, we analyze counts and percentages.

Why it matters: Poor data understanding often leads to poor models.

Real-world example: Before predicting salaries, check salary distribution and identify extreme values.

Interview Question: Why is EDA important? Answer: It helps understand data quality, distributions, missing values, and preprocessing needs.

Feature Scaling & Normalization

Machine learning algorithms can be affected when features have different ranges.

Standardization transforms data so that mean = 0 and standard deviation = 1.

Normalization scales values into a fixed range, usually 0 to 1.

Important for KNN, K-Means, PCA, SVM, and Neural Networks.

Tree-based models generally do not require scaling.

Interview Tip: If one feature ranges from 1-10 and another from 1-100000, distance-based algorithms may become biased.

Ordinal Encoding

Used when categories have a natural order.

Example: Low, Medium, High can become 0, 1, 2.

Advantages: Simple and efficient.

Disadvantage: Should not be used for categories without order.

Real-world example: Education level, customer satisfaction levels.

One-Hot Encoding

Used for nominal categories where no order exists.

Example: Red, Blue, Green become separate binary columns.

Advantage: Prevents false ordering.

Disadvantage: Can create many columns when categories are numerous.

Interview Tip: One-Hot Encoding is usually preferred for nominal data.

Column Transformer

Allows different preprocessing techniques for different columns.

Example: Scale numerical columns and One-Hot Encode categorical columns in the same workflow.

Benefits include cleaner code and reduced preprocessing errors.

Very common in production machine learning pipelines.

Machine Learning Pipelines

A pipeline combines preprocessing and model training into one workflow.

Benefits:

1. Cleaner code.
2. Prevents data leakage.
3. Easier deployment.
4. Easier experimentation.

Industry projects often rely heavily on pipelines.

Function Transformer

Used to apply mathematical transformations.

Examples:

- Log Transform
- Square Root Transform
- Reciprocal Transform

These transformations help make skewed distributions more normal and improve model performance.

Handling Date and Time Variables

Dates contain valuable information.

Features that can be extracted:

- Day
- Month
- Year
- Weekday
- Quarter
- Time Difference

Example: Customer purchase date can help predict buying behavior.

Missing Values

Missing values are extremely common in real-world datasets.

Methods:

- Complete Case Analysis
- Mean/Median Imputation
- Most Frequent Imputation
- Missing Indicator
- KNN Imputer
- MICE (Iterative Imputer)

Choosing the correct method depends on data type and business context.

Interview Question: Why not always drop rows? Answer: It may cause loss of valuable information.

Outliers & Feature Engineering

Outliers are extreme observations that differ from most data points.

Detection Methods:

- Boxplot
- IQR Method
- Z-Score

Feature Engineering is the process of creating better features from existing data.

Example:

Date of Birth -> Age

Full Name -> First Name + Last Name

Good feature engineering can improve model performance more than changing algorithms.